SUMMARY · MK 2 FINANCIAL MANAGEMENT & STRATEGY · WEEK 10

Risk and Refinements in Capital Budgeting

Last week every project was discounted at one rate (the cost of capital), as if all were equally risky. This week drops that assumption. Risk in capital budgeting is uncertainty about a project's cash flows. Three behavioural approaches give a "feel" for risk: the breakeven cash inflow (the minimum inflow for NPV ≥ 0 and the chance of reaching it), scenario analysis (pessimistic, most likely, optimistic NPV) and simulation (the full NPV distribution). Cross-border projects add exchange-rate and political risk, plus tax, transfer-pricing and strategic issues. The main way to put risk into NPV is the risk-adjusted discount rate (RADR), via the CAPM, a risk index or risk classes; with RADRs, Bennett's ranking flips from A to B. The refinements cover unequal lives (annualised NPV), real options (abandonment, flexibility, growth, timing → strategic NPV) and capital rationing (choose the set with the highest total NPV within the budget).

Material

1 Why risk matters: dropping the "one rate for all" assumption

So far (Chapters 10–11)

All projects are assumed equally risky; accepting one doesn't change the firm's risk; everything is discounted at the cost of capital.

vs

In reality (Chapter 12)

Risk differs by project; a big project can raise or lower the firm's risk; the decision must reflect each project's risk.

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Risk in capital budgeting = uncertainty about a project's cash
flows: how far they can vary from what we expect.
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Running case: Bennett Company (from Week 9). Cost of capital 10%:

Project	Yr 0	Yr 1	Yr 2	Yr 3	Yr 4	Yr 5	NPV @1(

A	−420,000	140,000	140,000	140,000	140,000	140,000	**$11**
B	−450,000	280,000	120,000	100,000	100,000	100,000	**$10**

At one rate, A wins narrowly ($1,466). The question: are A and B really equally risky? The answer changes three times this week.

Fact

In a survey of more than 1,000 US CEOs, almost 80% use NPV, yet nearly half say "gut feel" also matters (Graham & Harvey, 2015). This week's approaches make that "feel" more measurable.

2 Behavioural approach 1: the breakeven cash inflow

The **breakeven cash inflow** = the minimum annual inflow for a project to be acceptable (NPV ≥ 0). Method: find the cash flow that makes NPV exactly zero.

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NPV = CF × [1 - (1 + r)^-n] ÷ r - CF₀ = 0 → breakeven CF = CF₀ ÷ annuity factor(r, n)
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Treadwell Tire: projects T1 and T2 each cost $10,000, last 15 years, at a 10% cost of capital.

1. Annuity factor (10%, 15 yrs) = [1 − 1.10^−15] ÷ 0.10 = **7.606**
2. Breakeven CF = 10,000 ÷ 7.606 = **$1,315 a year**

Both need $1,315 a year. The next step is to judge the **probability** of reaching it:

	T1	T2
P(CF > $1,315)	100%	65%
Meaning	Certain to clear breakeven	35% chance of a negative NPV

So T1 looks safer than T2.

Limitation

The probabilities are often judgement calls. The method also says nothing about *how big* a loss could be.

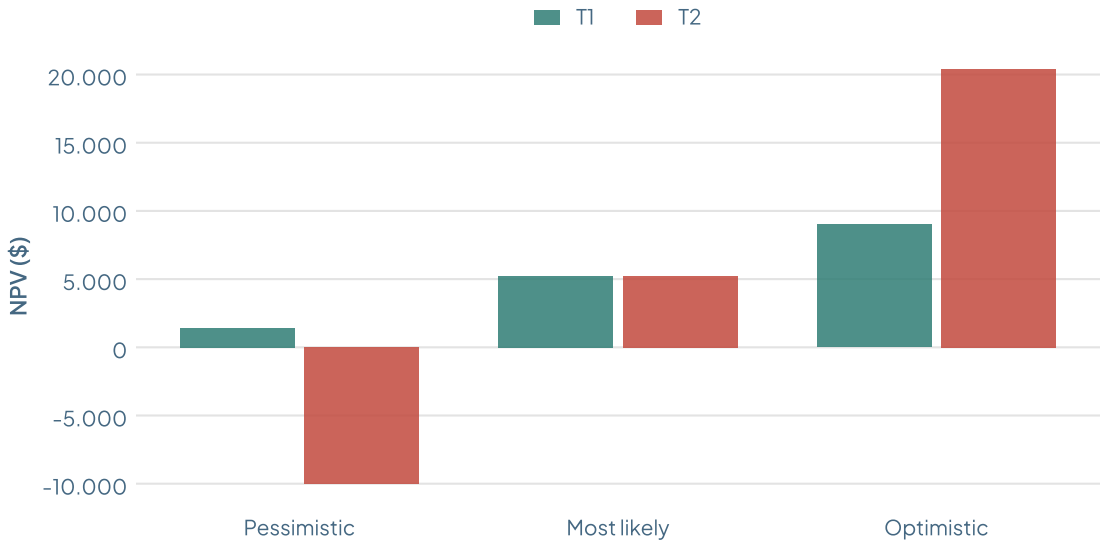
3 Behavioural approaches 2 and 3: scenario analysis and simulation

Scenario analysis computes NPV for pessimistic, most likely and optimistic cases. Treadwell ($10,000 cost, 15 years, 10%):

Scenario	T1 annual CF	T2 annual CF	T1 NPV	T2 NPV
Pessimistic	$1,500	$0	$1,409	−$10,000
Most likely	2,000	2,000	5,212	5,212
Optimistic	2,500	4,000	9,015	20,424
Range	$1,000	$4,000	**$7,606**	**$30,424**

How: NPV = CF × 7.606 − 10,000. E.g. T1 pessimistic: 1,500 × 7.606 − 10,000 = $1,409.

Treadwell NPV by scenario ($)



Same most-likely NPV ($5,212), but T2 can lose $10,000 or gain $20,424. T1's range is only $7,606.

Decision: same most-likely NPV ($5,212), but T1 has a far smaller range and no chance of loss. A risk-averse manager picks T1.

Simulation (e.g. Monte Carlo): give each uncertain input (price, volume, cost) a probability distribution, draw random values, compute NPV, repeat thousands of times. The result is the full NPV distribution: the expected NPV and the **probability of NPV < 0** (the share of negative outcomes = the chance of destroying value).

	Breakeven	Scenario analysis	Simulation
Question	Minimum CF for NPV ≥ 0	NPV in bad, base, good cases	Full NPV distribution
Output	One threshold + a probability	Three NPVs and a range	Expected NPV, P(NPV < 0)
Strength	Very simple	Easy to explain to a board	Most complete picture
Weakness	Probability is often a guess	Ignores how likely each case is	Needs data and a model

≡ **Applying it**

For a new toll road, the first input to stress-test is traffic volume (most uncertain and drives revenue). For a refinery, the gap between product and crude prices (the crack spread).

4 International risk

Exchange-rate risk

An unexpected change in the exchange rate cuts the value of the project's cash flows in the parent's currency. It arises when revenue is in one currency and costs or debt in another. Managed by **hedging**

vs

Political risk

A host government acts against foreign investors, or political turmoil endangers the investment: expropriation, sudden regulation, capital controls. Managed by deal structure, local partners and insurance.

and **matching currencies** (revenue and debt in the same currency).

The techniques stay the same (NPV, IRR); the cash flows and their risk must reflect these extra exposures.

Other issue	Explanation
Taxes	Only after-tax cash flows count, including taxes paid abroad and their effect on the parent's home tax bill.
Transfer pricing	Prices subsidiaries charge each other. Unless they reflect real costs, project cash flows are distorted.
Strategic motives	Investing for market access, to challenge a rival on its home turf, or to secure raw materials, even when the stand-alone NPV is weak.

Watch out

"Strategic" shouldn't become a reason to skip the numbers. Value the strategy explicitly (e.g. as a real option, see the refinements section).

Example

An Indonesian firm builds a plant in Vietnam with a USD loan, revenue in dong and parent accounts in rupiah. Three currencies: if the dong weakens against the USD, debt service gets heavier. The fix is local-currency borrowing or hedging.

5 The risk-adjusted discount rate (RADR)

1 · Adjust the cash flows

Cut each uncertain inflow to a level the firm is sure of (*certainty equivalent*). Highly subjective, less common.

vs

2 · Adjust the discount rate

Discount riskier projects at a higher rate: the **RADR**. Market-based, easy to apply, most common.

$$NPV = \Sigma\ CF_t \div (1 + RADR)^t - CF_0$$

The **RADR** = the return a project must earn to compensate the owners for its risk, keeping the share price at least unchanged.

A personal example: Talor's two stocks

	Stock M	Stock N
Price today	$1,000	$1,000
Annual dividend (5 years)	$80	$80
Expected sale price in year 5	$1,200	$1,500
Risk	Average	Considerably higher
RADR	11%	14%
Risk-adjusted NPV	$7.81	$53.70

Stock N is discounted at a higher rate yet still wins: its extra expected value more than pays for its extra risk.

RADR and the CAPM: project RADR = $r_f + \beta_project \times (r_m - r_f)$, i.e. the security market line (SML). Projects above the SML (IRR > required return → NPV > 0) are accepted; below it, rejected. The catch: real assets aren't traded in efficient markets, so project betas are hard to observe.

In practice: a risk index (Bennett). Management scores each project's risk (0 = risk-free); a higher score means a higher required return.

Risk index	0.0	0.4	0.8	1.0	1.2	1.6	2.0
RADR	6% (risk-free)	8%	10%	11% ← B	12%	14% ← A	18%

Bennett	Project A	Project B
Risk index	1.6	1.0
RADR	14%	11%
NPV @10% (no adjustment)	$110,710	$109,244
NPV @RADR	**$60,631**	**$97,984**

The ranking flips

The cash flows didn't change, only the rate. A is riskier, so it's discounted at 14% and its NPV falls by $50,079. B is discounted at 11% and now wins by $37,353. Lesson: one company-wide rate favours risky projects and penalises safe ones.

Risk classes (even simpler):

Class	Description	Typical project	RADR
I	Below-average risk	Routine replacement	8%
II	Average risk	Replacement or renewal of existing activities	10%
III	Above-average risk	Expansion of existing or similar activities	14%
IV	Highest risk	Expansion into new or unfamiliar activities	20%

Bennett: A → class III (14%), NPV $60,631. B → class II (10%), NPV $109,244. B is preferred. Risk classes are popular because managers think in rates of return, and a few classes are easy to estimate and apply.

Portfolio effects: should firms diversify? If a new business line lowers the firm's total risk, may it be accepted at a lower rate? (1) Investors can diversify themselves by holding many firms' shares; (2) so the market generally doesn't reward corporate diversification; (3) in practice firms use total project risk as an approximation. **Bottom line: judge each project at the rate matching its own risk, not the firm's.**

6 Refinement 1: unequal lives (annualised NPV)

AT Company, 10% cost of capital, two *mutually exclusive* projects:

Project	Yr 0	Yr 1	Yr 2	Yr 3	Yr 4	Yr 5	Yr 6
X (3 yrs)	−70,000	28,000	33,000	38,000			

Y (6 yrs)	−85,000	35,000	30,000	25,000	20,000	15,000	10,000

Y's NPV is higher, but Y also gives 3 more years of service, and X could be repeated after year 3. Comparing a 3-year NPV with a 6-year NPV isn't like-for-like.

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ANPV = NPV ÷ annuity factor(r, n) = NPV ÷ {[1 - (1 + r)^-n] ÷ r}
```

1. **Find each NPV** over its own life: X $11,277, Y $19,013.
2. **Convert to an annuity:** annuity factor at 10% for 3 years = 2.487; 6 years = 4.355. ANPV X = 11,277 ÷ 2.487 = **$4,535**; ANPV Y = 19,013 ÷ 4.355 = **$4,366**.
3. **Pick the highest ANPV:** X wins, $4,535 a year beats $4,366. The opposite of the NPV ranking.

Meaning

ANPV is "value per year of service". If X is repeated twice (6 years total), it is worth more than Y once.

7 Refinement 2: real options

Real options (strategic options) are flexibilities that let managers change a project's cash flows and risk after it starts. That flexibility has value.

Type	What management can do	Example
Abandonment	Stop the project early to limit losses	Close a mine if prices collapse
Flexibility	Switch inputs, outputs or processes	A plant that runs on gas or coal
Growth	Follow up with related projects later	A pilot plant that opens a new market
Timing	Choose when to invest	Wait for a permit or better prices

```
Strategic NPV = traditional NPV + value of real options
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Bennett, project B has two options:

Option	NPV if used	Probability	Expected value
Contract manufacturing (spare capacity, years 1–2)	$27,000	60%	$16,200
Computer-control sharing (two other machines)	$22,000	30%	$6,600
Value of real options			**$22,800**

Strategic NPV of B = 109,244 + 22,800 = **$132,044**. B now clearly beats A ($110,710), which has no real options.

> ! **Link to "strategic motives"**
> Real options are the way to value strategy explicitly, rather than calling a project "strategic" without numbers.

8 Refinement 3: capital rationing

Capital rationing happens when the budget is fixed (here $250,000) but there are more acceptable projects than can be funded. Cost of capital 10%.

Project	Investment	IRR	PV of inflows @10%	NPV
P2	$70,000	20%	$112,000	$42,000
P3	100,000	16%	145,000	45,000
P5	60,000	15%	79,000	19,000
P1	80,000	12%	100,000	20,000
P6	110,000	11%	126,500	16,500

P4	40,000	8%	36,000		−4,000 (IRR < 10%, reject)

IRR approach

Rank by IRR and fund down the list until the budget runs out: P2 + P3 + P5 = $230,000 used. P1 and P6 are acceptable but don't fit. Total NPV = $336,000 − $230,000 = **$106,000**.

vs

NPV approach

Find the combination with the highest total NPV that fits the budget: P2 + P3 + P1 = $250,000 (the whole budget). Total NPV = $357,000 − $250,000 = **$107,000**.

Rule

Under capital rationing, pick the set of projects with the **highest total NPV** that fits the budget. Unused money doesn't raise owners' wealth, so an IRR ranking doesn't guarantee the most value.

How to search: list the combinations that fit the budget, add their NPVs, pick the largest. Another example: P3 + P5 + P1 = $240,000, NPV $84,000 (smaller).

9 Wrap-up: which tool for which problem

If the problem is…	Use…	Decision rule
We need a feel for how risky the cash flows are	Breakeven, scenarios, simulation	Judge the chance and size of a loss
Projects differ in risk	RADR or risk classes	Accept if NPV at its own RADR > 0
Mutually exclusive projects with unequal lives	Annualised NPV (ANPV)	Pick the highest ANPV
The project has flexibility management can use	Strategic NPV (with real options)	Add the expected value of the options
The budget is fixed	NPV approach to capital rationing	Highest total NPV within the budget

Bennett: three answers

Bennett: NPV three ways ($)



At one rate A wins narrowly. Once risk and real options are counted, B wins.

	One rate (10%)	Risk-adjusted	With real options
A	$110,710	$60,631 (14%)	$110,710
B	$109,244	$97,984 (11%)	$132,044
Winner	A, narrowly	B	B, clearly

Discussion question

In your organisation, is every project judged at one hurdle rate? Which projects might be wrongly accepted (risky expansions) or wrongly rejected (safe replacements) as a result?

Key takeaways

1. Capital-budgeting risk = uncertainty about a project's cash flows. One rate for every project ignores risk differences.
2. Breakeven cash inflow = $CF_0 \div$ annuity factor; then judge the chance of reaching it (Treadwell: $1,315/yr; T1 100%, T2 65%).
3. Scenario analysis: pessimistic/most likely/optimistic NPV; a wide range = high risk (T1 $7,606 vs T2 $30,424).
4. Simulation gives the full NPV distribution: expected NPV and P(NPV < 0).
5. International risk: exchange rate (hedging, matching currencies) and political (structure, local partners, insurance); plus taxes, transfer pricing, strategic motives.
6. Two ways to include risk: adjust cash flows (certainty equivalent, subjective) or the discount rate (RADR, most common).
7. RADR via the CAPM/SML, a risk index or risk classes (I 8%, II 10%, III 14%, IV 20%).
8. Bennett with RADRs: A (14%) $60,631, B (11%) $97,984 → the ranking flips to B.
9. Judge each project at its own risk; investors can diversify themselves, so corporate diversification generally isn't rewarded.
10. Unequal lives: ANPV = NPV ÷ annuity factor; AT Company X $4,535 > Y $4,366 though Y's NPV is larger.
11. Real options (abandonment, flexibility, growth, timing): strategic NPV = NPV + option value; B becomes $132,044.
12. Capital rationing: choose the set with the highest total NPV within the budget (P2+P3+P1 $107,000 > P2+P3+P5 $106,000).

Case studies

Case & worked answers

Practice: a rooftop solar project at an SOE with RADR and a real option

An illustrative practice example, not from the slides. An SOE weighs two mutually exclusive projects, each costing IDR 100bn and lasting 5 years. The firm's cost of capital is 10%.

- **Project R** (replacing plant machinery, class II): cash flow IDR 28bn a year.
- **Project S** (rooftop solar sold to an industrial estate, a new business, class IV, RADR 20%): cash flow IDR 35bn a year.

1. **One rate of 10%** (annuity factor 3.791): NPV R = 28 × 3.791 – 100 = IDR 6.1bn; NPV S = 35 × 3.791 – 100 = **IDR 32.7bn**. S looks far better.
2. **RADR:** R at 10% stays IDR 6.1bn. S at 20% (factor 2.991): 35 × 2.991 – 100 = **IDR 4.7bn**. Now R is slightly better.
3. **S breakeven at 20%:** 100 ÷ 2.991 = IDR 33.4bn a year. If the power tariff falls 5% (cash flow 33.25), S's NPV turns negative. A thin safety margin.
4. **Real option:** S opens an option to expand to other industrial estates. At a 40% chance with an extra NPV of IDR 20bn, the option is worth 0.4 × 20 = IDR 8bn. Strategic NPV of S = 4.7 + 8 = **IDR 12.7bn** > R.

Lesson: one rate makes risky projects look too good; the RADR corrects that; a real option can flip it again, provided its value is calculated explicitly with a sensible probability.

Worked examples

Problem 1 **Easy** **Breakeven cash inflow**

A project costs $10,000, lasts 15 years, cost of capital 10%. What is the minimum annual cash flow for NPV ≥ 0? If the chance of beating it is 65%, what is the chance of a negative NPV?

1. `Breakeven CF = CF₀ ÷ {[1 - (1 + r)^-n] ÷ r}`
2. Annuity factor = [1 – 1.10^–15] ÷ 0.10 = 7.606
3. Breakeven CF = 10,000 ÷ 7.606 = $1,315
4. P(NPV < 0) = 1 – 65% = 35%

Answer: $1,315 a year; a 35% chance of negative NPV.

Problem 2 Easy **Scenario analysis**

For the same project (annuity factor 7.606, cost $10,000), T2 cash flows: pessimistic $0, most likely $2,000, optimistic $4,000. Compute each scenario's NPV and the range.

1. NPV = CF × 7.606 – 10,000
2. Pessimistic: 0 × 7.606 – 10,000 = –$10,000
3. Most likely: 2,000 × 7.606 – 10,000 = $5,212
4. Optimistic: 4,000 × 7.606 – 10,000 = $20,424
5. Range = 20,424 – (–10,000) = $30,424

Answer: –$10,000 / $5,212 / $20,424; range $30,424 (far riskier than T1's $7,606).

Problem 3 Medium **NPV with a RADR**

Bennett project A: investment $420,000, cash flow $140,000 a year for 5 years. Risk index 1.6 → RADR 14%. Compute NPV at 10% and at the RADR.

1. `NPV = CF × [1 - (1 + RADR)^-n] ÷ RADR - CF₀`
2. Factor 10%, 5 yrs = [1 – 1.10^–5] ÷ 0.10 = 3.7908 → NPV = 140,000 × 3.7908 – 420,000 = $110,710
3. Factor 14%, 5 yrs = [1 – 1.14^–5] ÷ 0.14 = 3.4331 → NPV = 140,000 × 3.4331 – 420,000 = $60,631
4. Difference = 110,710 – 60,631 = $50,079 due to risk

Answer: NPV @10% $110,710; NPV @14% $60,631. Still acceptable, but loses to B ($97,984 @11%).

Problem 4 Medium **Annualised NPV (ANPV)**

X: –70,000; 28,000; 33,000; 38,000 (3 years). Y: –85,000; 35,000; 30,000; 25,000; 20,000; 15,000; 10,000 (6 years). Cost of capital 10%. Choose using ANPV.

1. NPV X = 28,000/1.1 + 33,000/1.1^2 + 38,000/1.1^3 – 70,000 = 25,455 + 27,273 + 28,550 – 70,000 = $11,277
2. NPV Y (same method, 6 years) = $19,013
3. Annuity factors: 3 yrs = 2.487; 6 yrs = 4.355
4. ANPV X = 11,277 ÷ 2.487 = $4,535; ANPV Y = 19,013 ÷ 4,355 = $4,366

Answer: Choose X (ANPV $4,535 > $4,366), even though Y's NPV is larger.

Problem 5 Medium **Strategic NPV with real options**

Project B's traditional NPV = $109,244. Option 1 (contract manufacturing): NPV $27,000, 60% chance. Option 2 (computer-control sharing): NPV $22,000, 30% chance. Compute strategic NPV and compare with A ($110,710, no options).

1. ```
   Strategic NPV = traditional NPV + Σ (option NPV × probability)
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2. Option 1 = 27,000 × 0.60 = $16,200
3. Option 2 = 22,000 × 0.30 = $6,600
4. Option value = 22,800 → strategic NPV = 109,244 + 22,800 = $132,044

Answer: $132,044 > A's $110,710 → choose B.
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Problem 6 Challenging **Capital rationing**

Budget $250,000, cost of capital 10%. Projects (investment; IRR; PV of inflows): P1 (80,000; 12%; 100,000), P2 (70,000; 20%; 112,000), P3 (100,000; 16%; 145,000), P4 (40,000; 8%; 36,000), P5 (60,000; 15%; 79,000), P6 (110,000; 11%; 126,500). Compare the IRR and NPV approaches.

1. Drop P4 (IRR 8% < 10%, NPV −4,000)
2. NPV: P1 20,000; P2 42,000; P3 45,000; P5 19,000; P6 16,500
3. IRR: rank P2 → P3 → P5 (total 230,000; P1 no longer fits). Total NPV = 42,000 + 45,000 + 19,000 = $106,000
4. NPV: try P2 + P3 + P1 = 250,000 (exact). Total NPV = 42,000 + 45,000 + 20,000 = $107,000
5. Other combinations that fit (e.g. P3 + P5 + P1 = $84,000; P2 + P6 + P5 = $77,500) are smaller

Answer: Choose P2 + P3 + P1: the whole budget, total NPV $107,000.

Practice questions

Try answering first. The answer key is below each question.

1. Risk in capital budgeting is defined as…

A. The chance the firm goes bankrupt
B. Uncertainty about a project's cash flows
C. Share-price volatility
D. The loan interest rate

Answer: B. How far cash flows can vary from what's expected.

2. Treadwell: cost $10,000, 15 years, 10% (annuity factor 7.606). The breakeven cash inflow is…

A. $667
B. $1,000
C. $1,315
D. $1,500

Answer: C. 10,000 ÷ 7.606 = $1,315 a year.

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3. A limitation of breakeven analysis is…

A. Too complex
B. Probabilities are often guesses and it says nothing about loss size
C. Needs computer simulation
D. Ignores the cost of capital

Answer: B. You know the chance of loss, not how big it is.

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4. In Treadwell's scenario analysis, a risk-averse manager picks T1 because…

A. Its most-likely NPV is higher
B. Its NPV range is far smaller with no chance of loss
C. It's cheaper
D. It lasts longer

Answer: B. Same most-likely NPV ($5,212); range T1 $7,606 vs T2 $30,424.

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5. A weakness of scenario analysis compared with simulation is…

A. Hard to explain to a board
B. It ignores how likely each case is
C. It needs thousands of calculations
D. It produces no NPV

Answer: B. Simulation gives the full distribution including P(NPV < 0).

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6. The main output of simulation is…

A. One cash-flow threshold
B. Three NPVs
C. An NPV distribution: expected NPV and P(NPV < 0)
D. The project IRR

Answer: C. Each uncertain input gets a distribution, then NPV is computed thousands of times.

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7. A project earns in one currency but borrows in another. This creates…

A. Political risk
B. Exchange-rate risk
C. Transfer-pricing risk
D. Only liquidity risk

Answer: B. Managed by hedging and matching currencies.

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8. Expropriation and sudden capital controls are examples of…

A. Exchange-rate risk
B. Political risk
C. Ordinary business risk
D. Transfer pricing

Answer: B. Managed by structure, local partners and insurance.

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9. The most common way to put risk into NPV is…

A. Certainty equivalents
B. The risk-adjusted discount rate (RADR)
C. Payback period
D. Raising the cash flows

Answer: B. RADR is market-based and easy; certainty equivalents are highly subjective.

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10. Talor's stocks: M (RADR 11%) NPV $7.81; N (RADR 14%) NPV $53.70. Conclusion…

A. Pick M because it's less risky
B. Pick N: its extra expected value more than pays for its extra risk
C. Reject both
D. They can't be compared

Answer: B. Both have NPV > 0, but N is higher after adjusting for risk.

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11. Under the CAPM, a project above the SML means…

A. IRR < required return → reject
B. IRR > required return → NPV > 0 → accept
C. Negative beta
D. A risk-free project

Answer: B. The catch: real-project betas are hard to observe.

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12. Bennett with a risk index: A (1.6 → 14%), B (1.0 → 11%). The result…

A. A still wins
B. B wins: $97,984 vs $60,631
C. Both rejected
D. Equal NPVs

Answer: B. The cash flows didn't change, only the rate. The ranking flips.

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13. Expansion into new or unfamiliar activities is risk class…

A. I (8%)
B. II (10%)
C. III (14%)
D. IV (20%)

Answer: D. Expanding existing/similar activities = III; new/unfamiliar = IV.

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14. Why shouldn't a firm accept a project at a lower rate just because it diversifies the firm?

A. Diversification is banned
B. Investors can diversify themselves, so the market generally doesn't reward it
C. Diversification always raises risk
D. Diversifying projects have no cash flows

Answer: B. Judge each project at a rate matching its own risk.

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15. AT Company: NPV X $11,277 (3 yrs), NPV Y $19,013 (6 yrs). Annuity factors at 10%: 2.487 and 4.355. The right choice…

A. Y, higher NPV
B. X, because ANPV $4,535 > $4,366
C. Y, longer life
D. They're equal

Answer: B. For unequal lives, compare value per year of service (ANPV).

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16. A plant that can run on gas or coal is an example of which real option?

A. Abandonment
B. Flexibility
C. Growth
D. Timing

Answer: B. Flexibility = can switch inputs, outputs or processes.

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17. Waiting for a permit or better prices before investing is which option?

A. Timing
B. Growth
C. Abandonment
D. Flexibility

Answer: A. Timing = choosing when to invest.

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18. Bennett B: NPV $109,244; option 1: $27,000 × 60%; option 2: $22,000 × 30%. B's strategic NPV…

A. $109,244
B. $122,444
C. $132,044
D. $158,244

Answer: C. 16,200 + 6,600 = 22,800; 109,244 + 22,800 = 132,044.

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19. Under capital rationing, the right rule is…

A. Rank by IRR and fund from the top
B. Pick the set with the highest total NPV that fits the budget
C. Pick the cheapest projects
D. Pick the fastest payback

Answer: B. P2+P3+P1 ($107,000) beats the IRR pick P2+P3+P5 ($106,000).

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20. Why can the IRR approach fail under capital rationing?

A. IRR is always wrong
B. Budget can be left unused, and idle money doesn't add owner wealth
C. IRR can't be computed
D. IRR ignores cash flows

Answer: B. The IRR pick uses only $230,000 of $250,000.

Sources

- P3MD MK2 Week 10 — Risk and Refinements in Capital Budgeting (Gitman & Zutter, Principles of Managerial Finance, Ch. 12)
- Graham, J. & Harvey, C. (2015) — CEO survey on NPV and gut feel